

Branden Palmer

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Education

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| University of Texas at Austin <i>Austin, TX, USA</i> | Jan 2026 – May 2029 |
| <ul style="list-style-type: none">• M.S. in Computer Science | |
| Arizona State University <i>Tempe, AZ, USA</i> | Aug 2020 – Dec 2025 |
| <ul style="list-style-type: none">• B.S. in Computer Science Summa Cum Laude• GPA: 3.83 Dean's List Spring 2024, Fall 2024, Spring 2025, Fall 2025 | |

Projects

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| Job Sourcerer Video Link <i>A centralized job application platform</i> | Jan 2025 – Present |
| Machine Learning PostgreSQL React SBERT TailwindCSS TypeScript | |
| <ul style="list-style-type: none">• Grew user base to 17 active users with only 3 uninstalls, demonstrating strong product-market fit for job search automation• Setup CD/CI pipeline utilizing GitHub Actions, enabling early regression detection and single click deployment to Firefox and Chrome Extension stores• Handcrafted state-of-the-art client side ATS algorithm utilizing cosine of similarity and SBERT semantic matching, achieved resume to job description matching within 5% of paid service at zero cost• Implemented browser extension architecture using background, content, and popup scripts to extract job data from third-party sites (e.g., Handshake) and persist structured records locally• Developed a client-side data layer using Drizzle ORM with a PGLite-backed database to manage jobs, resumes, and derived analytics without a backend dependency | |
| Autonomous Driving Trajectory Prediction <i>University of Texas at Austin</i> <i>Deep Learning Project</i> | Jan 2026 – Apr 2026 |
| Python PyTorch Transformers Tensorboard CNNs CUDA NumPy Matplotlib | |
| <ul style="list-style-type: none">• Built and trained deep learning models in PyTorch to predict future vehicle trajectories for autonomous driving from lane geometry and raw image inputs• Implemented and compared an MLP planner, Transformer-based planner, and CNN planner for waypoint prediction on the SuperTuxKart Drive Dataset• Developed model-agnostic training pipelines from scratch, including data loading, optimization, checkpointing, and offline evaluation• Evaluated planning accuracy using longitudinal and lateral trajectory error, comparing structured-input and end-to-end vision-based driving approaches | |
| Applied Stylometry Video <i>Arizona State University</i> <i>Data Science Report</i> | Nov 2025 – Dec 2025 |
| Python pandas NumPy scikit-learn NLP TF-IDF PCA SciPy Matplotlib Plotly | |
| <ul style="list-style-type: none">• Applied stylometry and NLP techniques to analyze authorship signals across multiple text corpora using TF-IDF, character n-grams, and sentence-level features• Built an end-to-end text analysis pipeline in Python, including HTML parsing, text normalization, feature extraction, and exploratory visualization• Reduced high-dimensional linguistic feature spaces using PCA and visualized clustering behavior to compare content-based vs style-based representations• Evaluated methodological limitations of stylometry on short-form texts, demonstrating how data volume and shared background can confound authorship attribution | |

Open Source Contributions

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| G8 Medical Technology Link <i>Medical App</i> | Jan 2024 – Apr 2024 |
| Java JavaFX SQLite | |
| <ul style="list-style-type: none">• Broke six-person team into three feature focused subteams, reduced crosstalk, increased dev momentum• Performed 37 merges, instrumental in synchronizing six-person team• Delivered all required specifications on time | |
| Super Productivity Link <i>Productivity App</i> | Apr 2022 – Jan 2024 |
| SCSS Angular.js | |
| <ul style="list-style-type: none">• ID'd regression bug by utilizing git bisect• Corrected UI issue by isolating problematic SCSS, resulting change merged into production branch | |

Other Experience

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| Senior Aircraft Technician <i>Lockheed Martin, Balad AB, Iraq</i> | Dec 2017 – Jan 2020 |
| <i>Held active secret security clearance</i> | |